

Peng Zhilin

Nanjing University of Aeronautic and Astronautic | peng.zhilin@nuaa.edu.cn | 18089580048

wechat:spacetraveler1 | <https://pengzhilinnuaa.github.io/>

Education

Nanjing University of Aeronautic and Astronautic, Aircraft Design and Engineering Sept 2023 – Now

- GPA: 3.8/5.0 (Latest Transcript)
- **Coursework:** Aircraft Aerodynamics, Fluid Mechanics, Engineering Thermodynamics, C++ Programming

Papers

Physics-Constrained Invertible Neural Networks for Airfoil Optimization Model at Low Reynolds Number Jan 2026-Jun 2026

Zhilin Peng, Haochen Shi, ZhanZhan Liu, Zhaolin Chen

ICAS2026 Accept

History-Guided Iterative Visual Reasoning with Self-Correction Dec 2025-Jan 2026

Xinglong Yang, *Zhilin Peng*, ZhanZhan Liu, Haochen Shi, Sheng-Jun Huang

ACMMM2026 Under Reviewed

Projects

Physics-Constrained Invertible Neural Networks for Low-Reynolds-Number Airfoil Optimization Mar 2025 – Mar 2026

- Proposed an invertible neural network model based on the FreIA (Framework for Easily Inverting Architectures) structure for low-Reynolds-number flows, constructing a bidirectional mapping between airfoil parameters and drag coefficients that enables high-accuracy forward prediction and inverse airfoil optimization to support aerodynamic inverse design tasks.
- Individually responsible for all programming and implementation; the model has already achieved bidirectional mapping between airfoil geometric parameters and the lift-to-drag ratio. Currently investigating airfoil parameterization methods and Physics-Informed Neural Networks (PINNs) to establish mappings for airfoil geometric points and to incorporate physical constraints into the model.

• Tools Used: Python

Intelligent Board Game Robot System Based on Four-Axis Gantry Slide Sept 2025 – Jun 2026

- Designed and implemented a chess-playing robot system employing a gantry slide structure. A Raspberry Pi (deploying YOLO and Katago) serves as the upper computer, while an STM32 microcontroller acts as the lower computer, driving stepper motors to perform piece sorting, placement, and gameplay.
- As the project Leader, reproduced AlphaGo's code and completed model training; due to the limited computational capacity of the Raspberry Pi, migrated to the open-source Katago API and finished deployment, fine-tuning, and parameter adjustment. Designed mechanical parts and performed 3D modeling (SolidWorks).
- Second Prize, Chess-Piece Sorting Track, China Engineering Robot Competition.
- National College Student Innovation and Entrepreneurship Training Program
- Tools Used: Python, SOLIDWORK, AutoCad

Thermally Powered Car Jan 2025 – Aug 2025

- Designed and fabricated a vehicle powered by temperature difference, integrating a thermoelectric generation system with a purely mechanical cam steering mechanism. Realized thermal-to-electrical energy conversion and autonomous track tracing without electronic control. Through systematic thermo-electro-mechanical coordination, the vehicle successfully passed all checkpoints on the "Long March" track and delivered voice broadcasts.
- Responsible for hardware PCB design (JLCPCB, Multisim) and the development of capacitor charging boost modules employing chips such as PW4040 and MT3608.

- First Prize, Thermally Powered Car Category, National University Engineering Practice and Innovation Competition
- Grand Prize (Champion), Thermally Powered Car Category, Jiangsu Provincial University Engineering Practice and Innovation Competition
- Tools Used: LichuangEDA, NI Multisim

Technologies

Languages: Python, C++, Matlab, Latex

Technologies: AutoCad, CATIA, SOLIDWORKS, Analysis, LichuangEDA, Origin, Laser Cutting, 3DPrint

Experience

Executive President, Students' Association for Science and Technology of NUAA

Jul 2025 – Jul 2026